

## Complete System Bill of Materials

All prices verified against vendor listings and actual purchases, June 2026. Raspberry Pi and all SSD/NAND storage reflect the current industry memory shortage (SSDs up 2-4× from 2025) and are expected to decline in 2027-2028. Icom frequently runs instant coupons — check [hamradio.com/special\\_offers](https://hamradio.com/special_offers) before purchase.

| ITEM                                           | DETAIL                                                                                | QTY | UNIT  | EXT            |
|------------------------------------------------|---------------------------------------------------------------------------------------|-----|-------|----------------|
| <b>FieldCommand Server (192.168.50.1)</b>      |                                                                                       |     |       |                |
| <b>Raspberry Pi 5 — 16 GB</b>                  | Model B, 16 GB RAM — main application server                                          | 1   | \$305 | \$305          |
| <b>Pironman 5-MAX tower</b>                    | SunFounder — dual NVMe RAID 0/1, OLED, tower cooler, RGB                              | 1   | \$95  | \$95           |
| <b>1 TB NVMe SSD</b>                           | 2× M.2 2280 PCIe — RAID 1 mirror (NAND shortage pricing — was ~\$75 in 2025)          | 2   | \$180 | \$360          |
| <b>Pi 27W USB-C PSU</b>                        | Official Raspberry Pi 27W                                                             | 1   | \$14  | \$14           |
| <b>MicroSD 32 GB</b>                           | Boot media / recovery                                                                 | 1   | \$10  | \$10           |
|                                                | <b>FieldCommand Server (192.168.50.1) subtotal</b>                                    |     |       | <b>\$784</b>   |
| <b>44Net Gateway (192.168.50.2)</b>            |                                                                                       |     |       |                |
| <b>Raspberry Pi 5 — 16 GB</b>                  | Dedicated AMPRNet WireGuard gateway (8 GB variant ~\$130 acceptable)                  | 1   | \$305 | \$305          |
| <b>Argon NEO 5 case</b>                        | With M.2 SATA expansion                                                               | 1   | \$35  | \$35           |
| <b>256 GB M.2 SATA SSD</b>                     | Gateway OS drive (NAND shortage pricing)                                              | 1   | \$45  | \$45           |
| <b>Pi 27W USB-C PSU</b>                        | Official Raspberry Pi 27W                                                             | 1   | \$14  | \$14           |
|                                                | <b>44Net Gateway (192.168.50.2) subtotal</b>                                          |     |       | <b>\$399</b>   |
| <b>Networking</b>                              |                                                                                       |     |       |                |
| <b>ASUS RT-BE58 Go</b>                         | Wi-Fi 7 travel router — 1 primary + 2 AiMesh nodes (street \$119; MSRP \$160)         | 3   | \$119 | \$357          |
| <b>UniFi Switch Lite 16 PoE</b>                | 16-port GbE, 8× PoE — central wired hub                                               | 1   | \$199 | \$199          |
| <b>CAT6 cables</b>                             | Assorted lengths                                                                      | 10  | \$3   | \$30           |
|                                                | <b>Networking subtotal</b>                                                            |     |       | <b>\$586</b>   |
| <b>WAN — Cellular + Satellite</b>              |                                                                                       |     |       |                |
| <b>Cellular modem/router — omnidirectional</b> | Omnidirectional 5G/LTE antenna+modem — dual-carrier cellular — primary WAN            | 1   | \$375 | \$375          |
| <b>Directional cellular antenna</b>            | Directional folding antenna — backup for weak-signal sites                            | 1   | \$425 | \$425          |
| <b>Spare PoE cable</b>                         | Outdoor-rated shielded CAT6                                                           | 1   | \$20  | \$20           |
| <b>Satellite internet dish kit</b>             | Standard dish — automatic failover WAN (price DROP from \$499; regional promos lower) | 1   | \$349 | \$349          |
| <b>Satellite Ethernet adapter</b>              | Official adapter                                                                      | 1   | \$25  | \$25           |
| <b>USB-to-Ethernet adapter</b>                 | GbE USB 3.0 — ASUS USB WAN port                                                       | 1   | \$18  | \$18           |
|                                                | <b>WAN — Cellular + Satellite subtotal</b>                                            |     |       | <b>\$1,212</b> |
| <b>Operator Workstations</b>                   |                                                                                       |     |       |                |
| <b>Raspberry Pi 500+</b>                       | Keyboard PC — 16 GB RAM, 256 GB NVMe, mechanical keyboard (Gateron KS-33 Blue)        | 4   | \$410 | \$1,640        |
| <b>Raspberry Pi 500+ Desktop Kit upgrade</b>   | Adds mouse, 27W PSU, HDMI cable per unit (buy as kit @ \$430 ea)                      | 4   | \$20  | \$80           |
| <b>Raspberry Pi Monitor</b>                    | 15.6" 1080p IPS, USB-C powered                                                        | 4   | \$100 | \$400          |
|                                                | <b>Operator Workstations subtotal</b>                                                 |     |       | <b>\$2,120</b> |
| <b>HF Station</b>                              |                                                                                       |     |       |                |

| ITEM                              | DETAIL                                                                                                                                                                                                                                                                                                                                                                                                                                        | QTY | UNIT    | EXT            |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---------|----------------|
| Icom IC-7300MK2                   | HF/50 MHz 100 W SDR — HDMI out, USB-C, LAN remote, CW decoder, improved RMDR — check Icom coupons at hamradio.com/special_offers                                                                                                                                                                                                                                                                                                              | 1   | \$1,500 | \$1,500        |
| Astron RS-35M-AP power supply     | 13.8 VDC linear regulated, 25 A cont / 35 A peak, dual meters, Anderson Powerpoles — qty 2: primary for IC-7300MK2 + redundant/spare for field ops                                                                                                                                                                                                                                                                                            | 2   | \$344   | \$688          |
| LDG IT-100                        | 125 W Icom autotuner — station-side, powered/controlled by transceiver                                                                                                                                                                                                                                                                                                                                                                        | 1   | \$180   | \$180          |
| USB cable                         | Radio to laptop, shielded                                                                                                                                                                                                                                                                                                                                                                                                                     | 1   | \$10    | \$10           |
| Windows laptop                    | Winlink Express + VARA HF + JS8Call — budget \$1,000 for capable unit with SSD and USB ports                                                                                                                                                                                                                                                                                                                                                  | 1   | \$1,000 | \$1,000        |
|                                   | <b>HF Station subtotal</b>                                                                                                                                                                                                                                                                                                                                                                                                                    |     |         | <b>\$3,378</b> |
| <b>PACTOR HF Modem</b>            |                                                                                                                                                                                                                                                                                                                                                                                                                                               |     |         |                |
| SCS PXdragon DR-9400              | Current-generation PACTOR-4 HF modem — replaces discontinued DR-7800/7400 series — PACTOR-1/-2/-3/-4, Packet Radio 1200/9600 baud, Robust Packet Radio — quad-core ARM v8 @ 1.8 GHz real-time Linux — 1-ppm TCXO — USB-C + Ethernet LAN standard — optional Wi-Fi/Bluetooth dongle — OLED display — Winlink, Airmail, RMS Express compatible — manufactured in Germany by SCS — US dealer warranty via SeaTech Systems (no overseas shipping) | 1   | \$2,099 | \$2,099        |
| IC-7300MK2 interface cable set    | Audio cable (mic/speaker) + CI-V control cable from modem DIN port to IC-7300MK2 — with ferrite filters — ordered from SeaTech Systems or DX Engineering at time of purchase                                                                                                                                                                                                                                                                  | 1   | \$60    | \$60           |
|                                   | <b>PACTOR HF Modem subtotal</b>                                                                                                                                                                                                                                                                                                                                                                                                               |     |         | <b>\$2,159</b> |
| <b>VHF / UHF Station</b>          |                                                                                                                                                                                                                                                                                                                                                                                                                                               |     |         |                |
| Yaesu FTM-510DR                   | 144/430 MHz dual band C4FM digital + FM — 55W VHF / 50W UHF — built-in GPS + APRS (1200/9600 baud) — Super-DX weak-signal mode — 2.4" color touch display — detachable panel — WiRES-X / Fusion digital — in-stock HRO (FTM-510DR ASP w/ DSP board \$749.95 if preferred)                                                                                                                                                                     | 1   | \$670   | \$670          |
| Comet CA-2X4SRNMO antenna         | Wideband 2m/70cm mobile vertical designed for SAR/EMCOMM — 140-160 / 435-465 MHz covers public-safety VHF (150-160 MHz) AND ham bands — 3.8/6.2 dBi — 150W — 40" fold-over — NMO connector                                                                                                                                                                                                                                                    | 1   | \$72    | \$72           |
| Comet CM-5NMO mag mount           | 3.25" heavy-duty magnetic base, NMO connector, RG-58 coax tail — vehicle roof or equipment case lid mount                                                                                                                                                                                                                                                                                                                                     | 1   | \$35    | \$35           |
| RG-8X coax jumper                 | Short jumper: CM-5NMO coax tail to FTM-510DR SO-239 antenna port                                                                                                                                                                                                                                                                                                                                                                              | 1   | \$15    | \$15           |
| Digirig Mobile                    | Rev 1.11 — USB digital modes interface — combines audio codec, serial CAT and PTT — single USB-C connection — works with Windows/Mac/Linux — enables FT8, JS8Call, SSTV, VARA FM, packet on VHF/UHF via the FTM-510DR                                                                                                                                                                                                                         | 1   | \$50    | \$50           |
| Mobilinkd TNC4                    | 1200/9600 baud KISS TNC — Bluetooth 4.2 (iOS + Android) + USB-C — AFSK/GFSK/4-FSK — DSP modem — TCXO precision timing — 900mAh battery (48 hr) — enables APRS KISS mode for YAAC/Direwolf on FieldCommand                                                                                                                                                                                                                                     | 1   | \$150   | \$150          |
| Digirig Yaesu FTM MiniDin10 cable | digirig.net SKU YAESU10MOBILE — 10-pin MiniDin to Digirig audio socket — explicitly listed for FTM-510DR — audio + PTT — enables VARA FM, Winlink, JS8Call, SSTV, APRS via laptop                                                                                                                                                                                                                                                             | 1   | \$20    | \$20           |
| Mobilinkd Yaesu FTM modem cable   | store.mobilinkd.com — 10-pin MiniDIN, 1.5m — fits FTM-510DR (same connector as FTM-100/200/300/400/500 series) — connects TNC4 to radio for APRS KISS mode                                                                                                                                                                                                                                                                                    | 1   | \$25    | \$25           |
| USB-C data cable for Digirig      | Standard USB-C cable supporting both power and data — NOT included with Digirig — required to connect Digirig to laptop                                                                                                                                                                                                                                                                                                                       | 1   | \$10    | \$10           |
|                                   | <b>VHF / UHF Station subtotal</b>                                                                                                                                                                                                                                                                                                                                                                                                             |     |         | <b>\$1,027</b> |
| <b>Transport Cases</b>            |                                                                                                                                                                                                                                                                                                                                                                                                                                               |     |         |                |

| ITEM                                                                                    | DETAIL                                                                                                                                                                                                                                                                                                                                               | QTY | UNIT            | EXT            |
|-----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----------------|----------------|
| <b>Harbor Freight Apache 4800 XL</b>                                                    | Weatherproof protective case — IP65 rated — interior 17-7/8 x 12-7/8 x 6-5/8 in — pick-and-pull foam insert — purge valve — latching clasps — polypropylene shell — comparable to Pelican 1550 at a fraction of the cost — available in black, tan, yellow, orange, green — regular \$59.99; coupons bring to \$49.99 or \$39.99 (Inside Track Club) | 6   | \$60            | \$360          |
| <b>Transport Cases subtotal</b>                                                         |                                                                                                                                                                                                                                                                                                                                                      |     |                 | <b>\$360</b>   |
| <b>Field Antenna System</b>                                                             |                                                                                                                                                                                                                                                                                                                                                      |     |                 |                |
| <b>Chameleon CHA MPAS 2.0</b>                                                           | Modular Portable Antenna System — 1.8-54 MHz — vertical/sloper/inverted-V/NVIS/man-pack — deploys under 2 min — hybrid match, MIL 2.0 + EXT 2.0 whips, spike mount, wire, coax, bag (USA250 25%-off sale thru mid-July 2026)                                                                                                                         | 1   | \$660           | \$660          |
| <b>Chameleon CHA URT1</b>                                                               | Universal Remote Tuner — weatherproof feedpoint tuner — 5-1,500 Ω, 125 W, 16,000 memories — power/control over coax                                                                                                                                                                                                                                  | 1   | \$390           | \$390          |
| <b>Chameleon antenna wire</b>                                                           | 300 ft tinned-copper Kevlar PTFE — long wires, NVIS, dipoles, radials                                                                                                                                                                                                                                                                                | 1   | \$300           | \$300          |
| <b>Wire winders</b>                                                                     | Heavy-duty, 75-100 ft capacity each                                                                                                                                                                                                                                                                                                                  | 4   | \$15            | \$60           |
| <b>Field Antenna System subtotal</b>                                                    |                                                                                                                                                                                                                                                                                                                                                      |     |                 | <b>\$1,410</b> |
| <b>Printing &amp; Accessories</b>                                                       |                                                                                                                                                                                                                                                                                                                                                      |     |                 |                |
| <b>Brother MFC-L3780CDW</b>                                                             | Color laser MFP — IAP packages, ICS forms, access cards (L3770 discontinued; successor model)                                                                                                                                                                                                                                                        | 1   | \$450           | \$450          |
| <b>LaCie Rugged 1 TB USB-C</b>                                                          | LaCie Rugged USB-C 1TB (STFR1000800) — drop resistant 4 ft, crush 1 ton, rain resistant — 130 MB/s — USB-C + USB-A adapter included — Seagate Rescue data recovery service included — ideal for field deployment backup                                                                                                                              | 1   | \$125           | \$125          |
| <b>USB flash 32 GB</b>                                                                  | Sneakernet transfer                                                                                                                                                                                                                                                                                                                                  | 1   | \$10            | \$10           |
| <b>USB GPS receiver</b>                                                                 | u-blox — time + position for dashboard                                                                                                                                                                                                                                                                                                               | 1   | \$35            | \$35           |
| <b>Printing &amp; Accessories subtotal</b>                                              |                                                                                                                                                                                                                                                                                                                                                      |     |                 | <b>\$590</b>   |
| Core infrastructure (server, gateway, network, WAN, workstations, printing/accessories) |                                                                                                                                                                                                                                                                                                                                                      |     | <b>\$5,691</b>  |                |
| HF station (IC-7300MK2, 2× Astron RS-35M-AP, LDG IT-100, cable, laptop)                 |                                                                                                                                                                                                                                                                                                                                                      |     | <b>\$3,378</b>  |                |
| VHF/UHF station (FTM-510DR, Comet CA-2X4SRNMO, mag mount, coax)                         |                                                                                                                                                                                                                                                                                                                                                      |     | <b>\$792</b>    |                |
| Transport cases (6× Harbor Freight Apache 4800 XL @ \$59.99 list)                       |                                                                                                                                                                                                                                                                                                                                                      |     | <b>\$360</b>    |                |
| Field antenna system (MPAS 2.0, URT1, wire, winders)                                    |                                                                                                                                                                                                                                                                                                                                                      |     | <b>\$1,410</b>  |                |
| <b>COMPLETE SYSTEM TOTAL</b>                                                            |                                                                                                                                                                                                                                                                                                                                                      |     | <b>\$14,075</b> |                |

**Recurring costs:** the cellular data plan — \$5/month standby, \$79-99/month during activations only · the satellite internet service — ~\$120/month or pausable plan · No software licenses, subscriptions, or per-seat fees.

**Pricing sources:** raspberrypi.com · R&L; Electronics · DX Engineering · GigaParts · Ham Radio Outlet · chameleonantenna.com — verified June 2026.